

Model Evaluation — ML-Based Traffic Speed Prediction

This document describes the evaluation of the trained models published in this repository ([paper_artifacts/trained_models_base](#)) accompanying the paper “Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability”.

Models

One multilayer perceptron per sensor direction (37 sensor directions; layer sizes 96-256-256-96-256, ReLU with dropout, ~142k parameters, RMSprop optimiser, MSE loss, float32). Each model ships with its fitted input (x) and output (y) RobustScaler as .pk1 files. Models were trained on the initial three-year window (1 December 2020 to 1 December 2023) of five-minute average-speed records from Greater Manchester drakewell sensors, using 15 engineered features (cyclic time encodings and event flags). These are the base models from which all warm-start re-training scenarios in the paper are initialised.

Evaluation protocol

Each sensor’s data is split chronologically 80/20; the final 20% of the training window serves as the held-out test set (approximately the last seven months of 2023 for the three-year window). Metrics: mean absolute error (MAE, mph), root mean squared error (RMSE, mph), and mean absolute percentage error (MAPE, %). For context, each row also reports two properties of the sensor’s full speed series: the coefficient of variation of speed over the three-year model window, 1 December 2020 to 1 December 2023 (Speed CoV, dimensionless) - i.e., the variability of exactly the data the model was trained and evaluated on; the within-time-slot CoV (standard deviation of speed after removing the sensor’s time-of-week mean profile, divided by mean speed, computed over calendar year 2023 - a characterisation window chosen for stability, which overlaps but is not identical to the held-out test period); and the congestion frequency (share of five-minute intervals below 50% of the sensor’s 85th-percentile free-flow speed, over the same three-year window). A complementary common-test-set evaluation (identical held-out periods for all training scenarios) is summarised in the last section.

Per-sensor results — 3-year training regime

All rows in this table come from the **3-year initial training regime** (trained on 1 December 2020 to 1 December 2023); the held-out test set is the final 20% of that window, i.e. approximately mid-2023 to December 2023 (exact boundary varies slightly per sensor with data completeness). These are the published models in `trained_models_base3y/`.

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Within- slot CoV	Congestion (%)	Test samples
1	1020 nw	1.95	2.92	8.02	0.124	0.099	0.50	61,132
2	1020 se	1.67	2.42	6.48	0.137	0.092	0.37	60,958
3	1056 e	1.36	1.92	6.06	0.086	0.078	0.10	59,752
4	1056 w	1.75	2.47	7.57	0.099	0.087	0.13	60,544
5	1063 e	1.45	2.50	5.76	0.097	0.082	0.41	61,213
6	1063 w	1.59	2.47	5.50	0.115	0.076	0.70	61,253
7	1071 ne	2.09	3.65	10.48	0.122	0.097	1.53	62,524
8	1071 sw	1.31	1.93	4.31	0.079	0.060	0.11	62,668
9	1163 nw	2.37	3.36	9.55	0.162	0.105	1.05	62,835
10	1163 se	2.97	4.74	16.52	0.184	0.140	5.00	62,843
11	1165 se	2.07	3.04	8.16	0.120	0.082	1.69	61,874
12	1173 n	1.83	2.63	6.38	0.521	0.091	28.79	58,707
13	1273 n	1.90	2.99	7.41	0.120	0.095	0.61	61,176
14	1276 e	1.56	2.24	6.85	0.157	0.091	1.14	60,057
15	1276 w	3.52	5.50	27.63	0.278	0.197	13.17	59,959
16	1385 n	2.10	3.24	11.30	0.178	0.149	5.20	58,161
17	1404 e	2.69	4.38	11.19	0.181	0.136	4.94	61,257
18	1404 w	2.60	3.93	10.09	0.191	0.120	5.61	60,197
19	1407 n	1.90	3.20	10.03	0.128	0.101	1.14	62,645
20	1407 s	2.14	3.12	9.71	0.169	0.109	3.67	62,573

#	Sensor	MAE (mph)	RMSE (mph)	MAPE (%)	Speed CoV	Within- slot CoV	Congestion (%)	Test samples
21	1413 ne	1.58	2.27	6.06	0.099	0.076	0.26	62,277
22	1413 sw	1.64	2.25	5.92	0.095	0.076	0.06	62,174
23	1414 ne	1.54	2.13	6.18	0.117	0.081	0.34	60,300
24	1414 sw	1.41	1.98	5.66	0.110	0.072	0.23	60,317
25	1417 s	1.47	2.08	7.07	0.111	0.091	0.21	62,237
26	1418 e	1.92	3.28	9.09	0.116	0.096	1.49	55,437
27	1418 w	1.74	3.06	7.90	0.091	0.090	0.53	55,694
28	1420 nw	2.34	3.59	10.31	0.158	0.109	3.46	62,082
29	1420 se	1.91	2.90	8.64	0.124	0.106	0.76	61,899
30	1425 e	1.80	2.48	3.36	0.066	0.048	0.13	62,208
31	1425 w	2.16	4.38	6.25	0.073	0.074	0.34	62,376
32	1426 e	2.26	3.25	4.32	0.072	0.066	0.10	61,908
33	1426 w	3.75	6.26	12.68	0.176	0.121	3.50	62,685
34	1428 e	3.88	7.14	12.68	0.155	0.113	4.32	61,259
35	1428 w	2.17	3.25	4.12	0.066	0.060	0.06	61,158
36	1429 e	2.02	2.81	7.04	0.107	0.088	0.07	61,680
37	1429 w	2.00	2.90	7.04	0.108	0.095	0.24	60,874

Aggregate: mean MAE 2.07 mph (median 1.92); mean MAPE 8.47% (median 7.41%). The highest-error sensor directions are 1276 west (MAPE 27.6%) and 1163 southeast (16.5%).

Per-sensor error is well explained by the variability columns of the table, and the figures below carry the argument. Against the Speed CoV and congestion frequency columns, MAPE correlates at Spearman $\rho = 0.78$ and 0.79 respectively; against the within-slot CoV the correlation rises to $\rho = 0.94$, visible as the tight diagonal in the right panel of the figure below. The left panel ranks all 37 sensors by within-slot CoV: 1276 west stands clearly first, matching its first place on error, while 1173 north - the sensor with the highest Speed CoV (0.521) - sits mid-pack on both. The discrepancy between its two CoV columns is a historical level shift (yearly mean speed 8.4, 14.3, 47.4, 46.8 km/h for 2020-2023, entirely inside the training window), which inflates the Speed CoV but is absorbed by the model's year feature; in its stable regime the road is highly predictable.

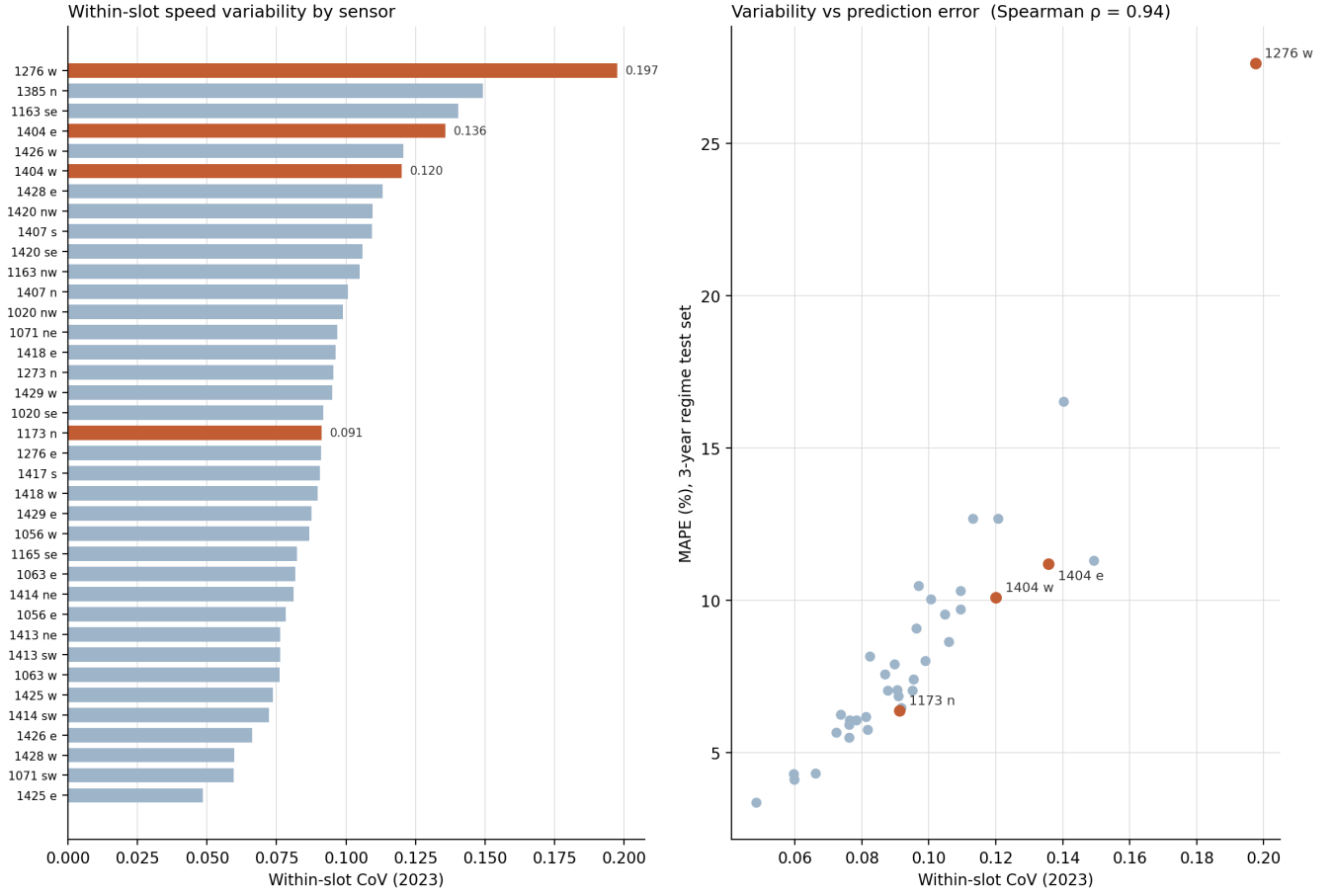


Figure 1: Left: within-slot CoV (2023) per sensor, sorted; highlighted bars are the sensors discussed in the text. Right: within-slot CoV against held-out MAPE under the 3-year regime (Spearman $\rho = 0.94$).

The figure below shows the mechanism directly. With all weeks of 2023 folded onto one Monday-Sunday axis, 1276 west's percentile band balloons through every daytime: the same clock time spans near-standstill to free flow depending on the week, so a calendar-feature model - which must predict a single value per slot - is bounded by that band width (a time-of-week average alone still leaves 22% MAPE there). On 1425 east, the narrow ribbon explains its 3.4% MAPE; 1173 north's 2023 band (middle panel) is nearly as narrow, confirming its predictability despite the Speed CoV. In short, prediction difficulty is driven almost entirely by day-to-day spread at the same time-of-week, which is exactly what the within-slot CoV column measures.

Speed fluctuation at the same time-of-week (2023)

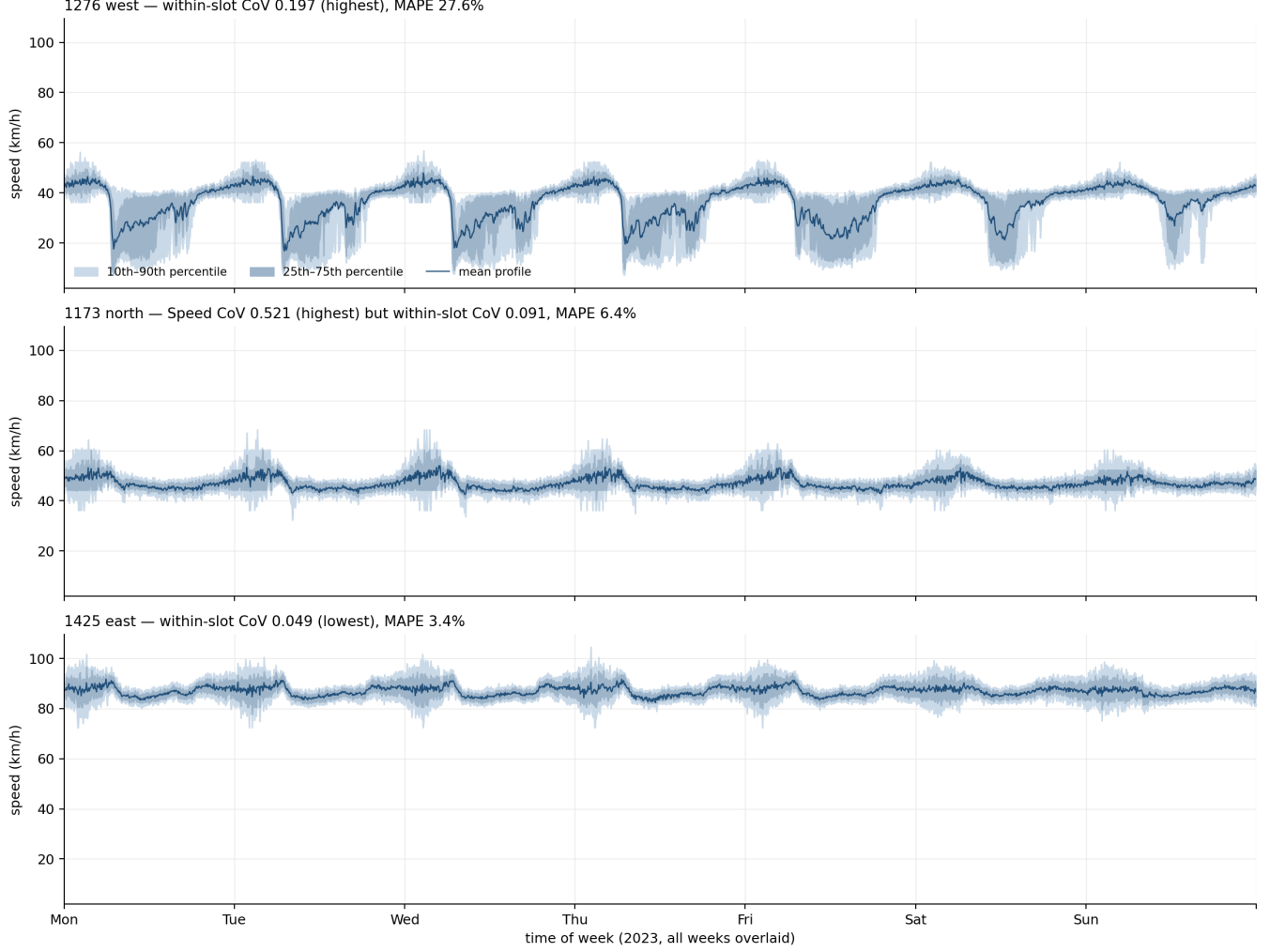


Figure 2: The within-slot spread seen directly: all weeks of 2023 folded onto one Monday-Sunday axis at 5-minute resolution. At each slot, the line is the mean across the ~52 weeks and the bands the 25th-75th and 10th-90th percentiles of those same weeks. On 1276 west (top) the same clock time spans near-standstill to free flow depending on the week. 1173 north (middle) shows why the Speed CoV misleads: within 2023 its band is a narrow ribbon despite the corpus-highest Speed CoV, which stems from pre-2022 level shifts. 1425 east (bottom, lowest within-slot CoV) barely fluctuates at all. A calendar-feature model predicts one value per slot, so the band width bounds its achievable accuracy.

Note that the error ranking is period-dependent: on the 2024-2025 common-test-set windows (last section), platform 1404 becomes the dominant outlier (MAPE up to 86.5%). The sensor data shows why: both directions slowed progressively from ~57 km/h (August 2024) to ~33 km/h (January 2025) with variance roughly tripling - a months-long, bidirectional degradation most likely due to long-term roadworks or road alteration at the site - so models trained before the change are evaluated against a changed road. Recurrent December dips are visible in earlier years too (2021-2023) but recovered each January; the 2024-2025 episode differs in onset (September, months before the seasonal dip), depth (10th-percentile speeds near standstill), and persistence (no January recovery by the end of the data).

Platform 1404: site-level regime change in both directions

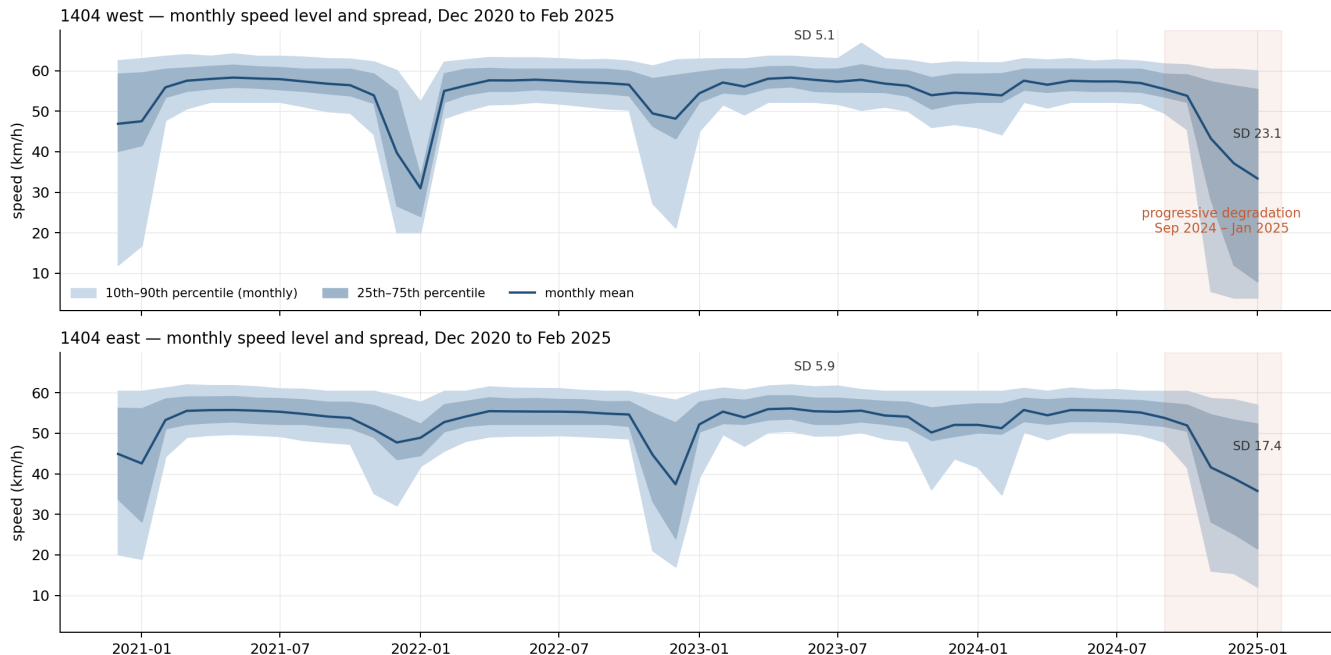


Figure 3: Platform 1404, both directions: monthly mean speed with 25th-75th and 10th-90th percentile bands over the full export (Dec 2020 - Feb 2025). Earlier December dips recover each January; the shaded September 2024 - January 2025 decline is deeper, starts months earlier, and shows no recovery, with the monthly standard deviation (SD, annotated on the chart) rising from ~5 to 17-23 km/h - a site-level regime change rather than seasonality.

Common-test-set scenario comparison

All training scenarios of the paper (3-year base, 4.2-year cold start, and progressive warm-start re-training windows PW3/PW6/PW12) were additionally evaluated on identical held-out periods. Mean MAE (mph) per evaluation window:

Scenario	Dec 2024 - Feb 2025	Oct 2024 - Feb 2025	May 2024 - Feb 2025
base (3y)	3.09	2.83	2.44
cold start (4.2y)	3.02	2.71	2.32
re-train PW3	2.91	2.68	2.35
re-train PW6	2.89	2.64	2.31
re-train PW12	2.90	2.61	excluded*

*PW12 trains up to September 2024, overlapping the nine-month window.

Warm-start re-training improves on both cold-start baselines in every window; prediction accuracy is invariant to the training platform (across repeated runs and platforms, the standard deviation of aggregate MAPE is below 0.3 percentage points in over 90% of measured cells).